ZOO511P – PRINCIPLES OF SYSTEMATICS

Practical Exam Notes

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Main Group link: https://chat.whatsapp.com/DWXwJpXahbxFVQutey8JXN

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MCQs (Multiple Choice Questions)

1. Sponges belong to phylum **PORIFERA**.
2. Nematodes include **UNSEGMENTED, CYLINDRICAL WORMS**.
3. The class of Pacific Sea Nettle is **SCYPHOZOA**.
4. The largest phylum is **ARTHROPODA**.
5. Arthropods muscles are **STRIATED**.
6. Porifers are called **SPONGES**.
7. Rotifers are **PSEUDOCOELOMATE**.
8. Echinodermata's distinguished character is that they have **SPINES** on their body.
9. Arthropods means **JOINTED LEGS**.
10. Molluscs have **OPEN** circulatory system.
11. T-test is less reliable than **ANOVA if THERE ARE MORE THAN TWO SAMPLES**.
12. Example of phylum Cnidaria is **ALCYONIUM**.
13. Nematodes are also called **ROUND WORMS**.
14. The excretory organ in Mollusca is **METANEPHRIDIA**.
15. Analysis of Variance (ANOVA) is a **STATISTICAL TECHNIQUE THAT IS USED TO CHECK THE MEAN OF TWO OR MORE GROUPS THAT ARE SIGNIFICANTLY DIFFERENT FROM EACH OTHER.**
16. Phylum Echinodermata are specialized for **WATER VASCULAR SYSTEM**.
17. Tissues are not present in phylum **PORIFERA**.
18. Annelids are **COELOMATE**.
19. The crown of cilia is called **CORONA**.
20. Dichotomous Key is used to **IDENTIFY AND CATEGORIZE ANIMALS**.

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SHORT QUESTIONS

1. Write classes of Phylum Porifera with examples.

1. Class Calcarea (Calcispongiae)

These sponges possess a skeleton made of calcium carbonate spicules. They are mostly marine organisms and are generally small in size. ***Examples:*** Sycon and Leucosolenia.

2. Class Hexactinellida (Hyalospongiae)

Commonly known as glass sponges, members of this class have a skeleton composed of six-rayed siliceous spicules. They are usually found in deep marine waters. ***Examples:*** Euplectella and Hyalonema.

3. Class Demospongiae

This is the largest class of sponges. Their skeleton consists of spongin fibers, siliceous spicules, or both. Most species are marine, while a few inhabit freshwater environments. ***Examples:*** Spongilla and Euspongia.

2. Write short note on habitat of *Amynthas mekongianus*.

Habitat of *Amynthas mekongianus*

Amynthas mekongianus is a species of earthworm belonging to the family *Megascolecidae*. It is primarily found in moist terrestrial habitats where the soil is rich in organic matter. This earthworm commonly inhabits agricultural fields, gardens, forests, grasslands, and areas near riverbanks.

The species prefers loose, well-aerated soil with adequate moisture because such conditions facilitate burrowing and provide abundant decaying plant material for food. Like other earthworms, *Amynthas mekongianus* avoids extremely dry environments and direct sunlight, as its skin must remain moist for respiration.

This earthworm plays an important ecological role by improving soil structure, increasing aeration, enhancing water infiltration, and promoting the decomposition of organic matter, thereby contributing to soil fertility and ecosystem health.

3. Write distinguishing characters of phylum Mollusca.

Distinguishing Characters of Phylum Mollusca

1. Molluscs are soft-bodied, unsegmented, and bilaterally symmetrical animals.
2. They are triploblastic and coelomate organisms with organ-system level organization.
3. The body is typically divided into three regions: head, muscular foot, and visceral mass.
4. A specialized fold of body wall called the mantle surrounds the visceral mass and usually secretes a calcareous shell.
5. Most molluscs possess an external shell made of calcium carbonate, although it may be reduced or absent in some species.
6. The muscular foot is used for locomotion, burrowing, attachment, or capturing prey.
7. The digestive system is complete, with a mouth and anus. Many species possess a rasping feeding organ called the radula.
8. Respiration occurs through gills (ctenidia), lungs, or the body surface, depending on the habitat.
9. The circulatory system is generally open, with a heart and blood-filled spaces called sinuses. However, cephalopods have a closed circulatory system.
10. Excretion is carried out by a pair of nephridia (kidneys).
11. The nervous system consists of ganglia connected by nerve cords.
12. Reproduction is mainly sexual. Sexes may be separate or united in the same individual, and development often includes larval stages such as trochophore and veliger larvae.
13. Examples are Pila, Unio, Octopus, and Sepia.

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4. ANOVA test is a suitable technique for a sample. Give example with reference to the context.

ANOVA (Analysis of Variance) is a statistical technique used to compare the means of three or more groups to determine whether there is a significant difference among them. It is suitable when a researcher wants to test the effect of one factor on a continuous variable across multiple groups.

Example:

Suppose a biologist wants to study the effect of three different fertilizers (A, B, and C) on the growth of plants. Three groups of plants are treated with the respective fertilizers, and their heights are measured after a fixed period.

- ***Group 1:*** Fertilizer A
- ***Group 2:*** Fertilizer B
- ***Group 3:*** Fertilizer C

The researcher can use one-way ANOVA to compare the mean plant heights of the three groups. If the ANOVA result shows a significant difference, it indicates that at least one fertilizer affects plant growth differently from the others.

5. Name three classes of molluscs with examples.

1. Class Gastropoda

Gastropods are the largest class of molluscs. They usually possess a single coiled shell and move with the help of a muscular foot. ***Examples*****:** Pila and Helix.

2. Class Bivalvia (Pelecypoda)

Members of this class have a body enclosed within two shell valves. They lack a distinct head and radula. ***Examples*****:** Unio and Ostrea.

3. Class Cephalopoda

These are highly developed molluscs in which the foot is modified into arms or tentacles surrounding the mouth. They possess a well-developed nervous system and sense organs. ***Examples*****:** Octopus and Sepia.

6. Write a short note on habitat of nematodes.

Habitat of Nematodes

Nematodes, commonly known as roundworms, are among the most widely distributed animals on Earth. They occur in a great variety of habitats, including soil, freshwater, marine environments, and within the bodies of plants and animals.

Free-living nematodes are commonly found in moist soil, decaying organic matter, river sediments, lakes, ponds, and ocean floors, where they feed on bacteria, fungi, algae, and other microorganisms. Many species play an important role in nutrient recycling and soil fertility.

Parasitic nematodes inhabit the tissues, organs, or digestive tracts of plants and animals. Some species infect humans, while others parasitize livestock, insects, and crop plants, often causing diseases and economic losses.

Nematodes can survive in a wide range of environmental conditions, from tropical forests to deserts and from shallow waters to deep-sea sediments. Their adaptability and abundance make them one of the most successful groups of invertebrates.

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7. Write some assumptions of ANOVA.

Assumptions of ANOVA

ANOVA (Analysis of Variance) is based on several important assumptions:

1. ***Independence of Observations:*** The observations in each group must be independent of one another. The value of one observation should not influence another.

2. ***Normal Distribution:*** The data within each group should be approximately normally distributed.

3. ***Homogeneity of Variance:*** The variances of all groups being compared should be equal or nearly equal.

4. ***Random Sampling:*** The samples should be selected randomly from the population.

5. ***Continuous Dependent Variable:*** The variable being analyzed should be measured on an interval or ratio scale.

6. ***Mutually Exclusive Groups:*** Each observation should belong to only one group, and the groups should not overlap.

8. Name three classes of Cnidaria with examples.

1. Class Hydrozoa

These cnidarians are mostly marine, but some are freshwater. They often exist in both polyp and medusa forms, with polyp being dominant in many species. ***Examples*:** Hydra and Obelia.

2. Class Scyphozoa

These are known as true jellyfish. The medusa stage is dominant, and the polyp stage is usually reduced. ***Examples*:** Aurelia and Cyanea.

3. Class Anthozoa

These cnidarians exist only in the polyp form and do not have a medusa stage. They are mostly marine and include corals and sea anemones. ***Examples*:** Metridium and Corallium.

9. Name seven classes of Arthropods with examples.

1. Crustacea: Mostly aquatic arthropods with two pairs of antennae and gills for respiration. ***Examples*:** Palaemon and Cancer.

2. Insecta (Hexapoda): Largest group of arthropods with three body regions (head, thorax, abdomen) and three pairs of legs. ***Examples*:** Apis and Musca.

3. Arachnida: Have four pairs of legs and lack antennae; mostly terrestrial. ***Examples*:** Araneus and Scorpio.

4. Chilopoda: Commonly called centipedes; elongated body with one pair of legs per segment. ***Examples*:** Scolopendra.

5. Diplopoda: Known as millipedes; cylindrical body with two pairs of legs per segment. ***Examples*:** Julus.

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6. Merostomata: Ancient group of marine arthropods with a hard horseshoe-shaped carapace. ***Examples*:** Limulus.

7. Trilobita (Extinct group): Extinct marine arthropods with a three-lobed body plan. ***Examples*:** Fossil genera such as *Calymene* and *Paradoxides*.

10. Name three classes of Nematodes with examples.

1. Class Enoplea (Adenophorea group): These nematodes are mostly free-living or parasitic. They have a simple excretory system and smooth body cuticle. ***Examples*:** Trichinella spiralis and Trichuris trichiura.

2. Class Chromadorea (Secernentea group): These are the most common nematodes, found in soil, water, and as parasites of plants and animals. They possess phasmids and a more developed excretory system. ***Examples*:** Ascaris lumbricoides and Enterobius vermicularis.

3. Class Dorylaimea: These are mostly free-living soil nematodes, often feeding on microorganisms or other small organisms. They are characterized by a long, needle-like stylet. ***Examples*:** Dorylaimus and Xiphinema.

11. Write a short note of body structure of Rotifers.

Body Structure of Rotifers

Rotifers are microscopic, aquatic pseudocoelomate animals commonly found in freshwater environments, though some also occur in moist soil and marine habitats.

The body of rotifers is typically divided into three main regions: the head, trunk, and foot.

The head region bears a characteristic ciliated structure called the corona, which appears like a rotating wheel under the microscope. The corona helps in locomotion and also creates water currents that bring food particles toward the mouth.

The trunk is the main body region containing the digestive system and other internal organs. Rotifers have a complete digestive tract with a specialized muscular pharynx called the mastax, which contains hard jaw-like structures known as trophi used for grinding food.

The body is covered by a tough, flexible outer layer called the cuticle, which may form a protective case known as a lorica in some species.

The foot region is posterior and often bears adhesive glands and toes that help in temporary attachment to substrates.

Rotifers exhibit a pseudocoelom, and their body size is generally very small, usually ranging from 0.1 to 0.5 mm.

12. What is the mode of reproduction in Arthropods?

Mode of Reproduction in Arthropods

Arthropods show a sexual mode of reproduction, and most species have separate sexes (dioecious)**,** meaning male and female individuals are distinct. Fertilization is usually internal, especially in terrestrial arthropods like insects and arachnids, while in some aquatic forms it may be external or indirect. Many arthropods are oviparous, meaning they lay eggs, although some species may be ovoviviparous or rarely viviparous. Development in arthropods often involves metamorphosis, which may be:

- ***Complete metamorphosis*** (egg → larva → pupa → adult), as seen in butterflies and beetles
- ***Incomplete metamorphosis*** (egg → nymph → adult), as seen in grasshoppers and cockroaches

Some arthropods also show parthenogenesis, where development of offspring occurs from unfertilized eggs, for example in some insects like aphids.

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13. Name classes of Annelids with example of each class.

1. Class Polychaeta: These are mostly marine annelids. They have many setae (bristles) on parapodia used for movement and respiration. They are usually dioecious. ***Examples*:** Nereis and Arenicola.

2. Class Oligochaeta: These annelids have fewer setae and lack parapodia. They are mostly terrestrial or freshwater and are hermaphrodites. ***Examples*:** Pheretima and Tubifex.

3. Class Hirudinea: These are commonly known as leeches. They have a fixed number of segments, lack setae, and possess suckers for attachment. ***Examples*:** Hirudo and Haemadipsa.

14. Write a note on excretion in Rotifers.

Excretion in Rotifers

Excretion in rotifers is carried out mainly by a specialized protonephridial system, which helps in removing metabolic wastes and maintaining osmotic balance.

The excretory system consists of a pair of nephridia (flame cell-like structures) located on either side of the body. These nephridia are made up of fine tubules that end in terminal flame bulbs (solenocytes)**.** The beating of cilia or flagella in these cells creates a filtration current that draws waste fluids into the excretory tubules.

The collected waste fluid then passes through the tubules and is finally released into a bladder-like structure, which opens to the exterior through an excretory pore**.**

In addition to nitrogenous waste removal (mainly ammonia), the excretory system of rotifers plays an important role in osmoregulation, especially in freshwater environments where excess water continuously enters the body by osmosis.

This system is highly efficient for their microscopic size and helps rotifers survive in varying aquatic conditions.

15. Write distinguishing characters of Arthropods.

Distinguishing Characters of Phylum Arthropoda

1. Arthropods have a bilaterally symmetrical, segmented body divided into distinct regions such as head, thorax, and abdomen (or cephalothorax and abdomen).
2. The body is covered by a hard chitinous exoskeleton, which provides protection and support.
3. They possess jointed appendages, which are modified for walking, feeding, sensing, swimming, or reproduction.
4. Arthropods exhibit triploblastic and coelomate organization, with a reduced coelom called a haemocoel.
5. The circulatory system is open, with haemolymph circulating through body cavities.
6. Respiration occurs through different structures depending on the group, such as tracheae (insects), gills (crustaceans), book lungs (arachnids), or body surface.
7. The digestive system is complete, with specialized mouthparts adapted for different types of feeding.
8. Excretion is carried out by specialized organs such as Malpighian tubules, green glands, or coxal glands**.**
9. The nervous system is well developed, consisting of a brain and a double ventral nerve cord with segmental ganglia.
10. Arthropods show sexual reproduction, and most species are dioecious with internal or external fertilization.
11. Growth is accompanied by molting (ecdysis) due to the rigid exoskeleton.
12. They show diverse developmental patterns, often including metamorphosis.

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16. Write a short note on reproduction in Molluscs.

Reproduction in Molluscs

Molluscs show a sexual mode of reproduction. Most species are dioecious (having separate male and female individuals), although some, like certain gastropods, are hermaphrodites.

Fertilization may be external or internal, depending on the class and habitat. In aquatic molluscs such as many bivalves, fertilization is usually external, whereas in gastropods and cephalopods it is generally internal.

Development in molluscs often includes one or more larval stages. The first larval stage is typically the trochophore larva, which is free-swimming and ciliated. In many species, it is followed by a veliger larva, which shows early development of shell and foot structures.

Some molluscs exhibit direct development, where the young resemble miniature adults and no larval stage is present.

In cephalopods, reproduction is highly specialized, with males transferring sperm using a modified arm called the hectocotylus.

17. Why formalin is used as preservative?

Why Formalin is Used as a Preservative

Formalin is widely used as a preservative because it effectively prevents decomposition of biological specimens and maintains their structural integrity for a long period.

1. ***Prevents decomposition:*** Formalin kills bacteria, fungi, and other microorganisms that cause decay.
2. ***Protein fixation:*** It preserves tissues by denaturing and cross-linking proteins, which helps maintain the shape and structure of organisms.
3. ***Long-term preservation:*** Specimens stored in formalin can remain in good condition for years without significant degradation.
4. ***Maintains morphology:*** It preserves the external and internal features of organisms, making it useful for anatomical and zoological studies.
5. ***Cost-effective and widely available:*** Formalin is relatively inexpensive and easy to use in laboratories.
6. ***Soluble in water:*** It mixes easily with water, allowing preparation of different concentrations for preservation needs.

18. Write a short note on habitat of *Plexipus petersi*.

Habitat of *Plexippus petersi*

Plexippus petersi is a species of jumping spider belonging to the family Salticidae. It is widely distributed in tropical and subtropical regions, especially in Asia.

This spider is mainly found in terrestrial habitats such as gardens, agricultural fields, shrubs, grasses, and around human dwellings. It prefers warm environments with plenty of vegetation where it can easily hunt its prey.

Plexippus petersi is commonly seen on plant leaves, tree trunks, walls, and fences, where it actively moves in search of insects. Unlike web-building spiders, it does not rely on webs for capturing prey; instead, it is a diurnal and active hunter that uses its excellent vision and jumping ability.

It is adaptable to both natural and semi-urban environments, which allows it to survive in a wide range of ecological conditions.

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19. What are feeding and economic importance of *Ascaris lumbricoides*?

Feeding and Economic Importance of *Ascaris lumbricoides*

Feeding

Ascaris lumbricoides is a parasitic roundworm that lives in the small intestine of humans. It does not have specialized feeding structures for active predation. Instead, it feeds on semi-digested food (intestinal contents**)** of the host by absorbing nutrients directly through its body surface (cuticle). It mainly utilizes carbohydrates and other nutrients already digested by the host.

Economic Importance

1. ***Human health impact:*** It causes ascariasis, a common intestinal disease in humans, especially in areas with poor sanitation.
2. ***Malnutrition:*** Heavy infection leads to nutrient deficiency, weight loss, and stunted growth, particularly in children.
3. ***Intestinal blockage:*** Large numbers of worms may cause intestinal obstruction, which can be life-threatening.
4. ***Reduced productivity:*** Infected individuals often suffer from weakness and fatigue, leading to decreased work efficiency.
5. ***Economic burden:*** Treatment costs, healthcare expenses, and loss of productivity make it economically significant in affected regions.
6. ***Public health concern:*** Its eggs spread through contaminated food and water, making it an important parasite in sanitation and hygiene management programs.

20. Write characteristics of phylum Porifera.

Characteristics of Phylum Porifera

1. Poriferans are commonly known as sponges and are the simplest multicellular animals.
2. They are aquatic, mostly marine, with a few freshwater species.
3. Their body shows cellular level of organization; true tissues and organs are absent.
4. They are generally asymmetrical, though some may show radial symmetry.
5. The body is full of numerous pores called ostia, through which water enters, and a large opening called osculum, through which water exits.
6. They possess a canal system that helps in water circulation for food, gas exchange, and excretion.
7. The body is supported by a skeleton made of spicules (calcareous or siliceous) and/or spongin fibres**.**
8. Digestion is intracellular, carried out by specialized cells called choanocytes (collar cells).
9. They are filter feeders, feeding on microscopic organisms and organic particles suspended in water.
10. Respiration, excretion, and circulation occur by diffusion through water flow.
11. Reproduction occurs by both asexual methods (budding, fragmentation, gemmules) and sexual reproduction**.**
12. Most sponges are hermaphroditic, and fertilization is usually internal.
13. Larval forms are free-swimming, aiding in dispersal and colonization.

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21. Write characteristics of phylum Annelida.

Characteristics of Phylum Annelida

1. Annelids are bilaterally symmetrical, triploblastic, and coelomate animals.
2. Their body is elongated and segmented, with external and internal segmentation (metamerism).
3. They possess a true body cavity (coelom) lined by mesoderm.
4. The body is divided into head, body segments, and sometimes a tail region.
5. The body wall contains circular and longitudinal muscles, allowing efficient movement.
6. Locomotion is aided by setae (chaetae) in most annelids; parapodia are present in polychaetes.
7. The digestive system is complete, with mouth and anus, and shows regional specialization.
8. Respiration occurs through skin (cutaneous respiration) or gills in some aquatic forms.
9. The circulatory system is closed, with blood flowing through vessels.
10. Excretion is carried out by nephridia.
11. The nervous system consists of a brain (cerebral ganglia) and a ventral nerve cord with segmental ganglia.
12. Reproduction may be sexual or asexual, with most species showing sexual reproduction.
13. Some annelids are dioecious (e.g., polychaetes) while others are hermaphroditic (e.g., earthworms, leeches).
14. Development may be direct or indirect, often involving a trochophore larva in marine forms.

22. Write characteristics of rotifers.

Characteristics of Rotifers

1. Rotifers are microscopic, aquatic, pseudocoelomate animals commonly found in freshwater habitats.
2. Their body is bilaterally symmetrical, triploblastic, and unsegmented.
3. The body is divided into head, trunk, and foot regions.
4. The head bears a ciliated structure called the corona, which appears like a rotating wheel and is used for locomotion and feeding.
5. They possess a complete digestive system with a specialized muscular pharynx called the mastax, containing jaw-like structures known as trophi.
6. The body is covered by a flexible outer cuticle, and in some species forms a protective structure called a lorica.
7. The excretory system is of protonephridial type, mainly used for osmoregulation.
8. Respiration occurs through the body surface by diffusion.
9. They exhibit a pseudocoelom, which acts as a hydrostatic skeleton.
10. Nervous system is simple with a cerebral ganglion and longitudinal nerve cords.
11. Reproduction is mainly sexual, but parthenogenesis (asexual reproduction) is very common.
12. Males are often smaller and short-lived, and in some species may be absent.

23. Name three classes of Rotifers with examples.

Rotifers are generally classified into three main classes:

1. Class Bdelloidea

These are exclusively freshwater rotifers. They reproduce mainly by parthenogenesis and lack males in most species. ***Examples*:** Philodina and Rotaria.

2. Class Monogononta

This is the largest class of rotifers. They show both sexual reproduction and parthenogenesis and are found in freshwater as well as some marine habitats. ***Examples*:** Brachionus and Lecane.

3. Class Seisonidea

These are rare marine rotifers found as ectosymbionts on crustaceans. They reproduce sexually and lack parthenogenesis. ***Examples*:** Seison and Seison nebaliae.

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LONG QUESTIONS

1. Write characteristics of *Plexipus petersi*.

Characteristics of *Plexippus petersi*

1. *Plexippus petersi* is a jumping spider belonging to family Salticidae under class Arachnida.
2. It is a terrestrial, free-living and predatory spider commonly found in gardens, fields, and human dwellings.
3. The body is divided into two regions: cephalothorax (prosoma) and abdomen (opisthosoma).
4. It has four pairs of jointed legs adapted for walking and jumping.
5. It possesses large, well-developed anterior median eyes, providing excellent vision for detecting prey.
6. Unlike web-building spiders, it is a diurnal active hunter that captures prey by jumping rather than using webs.
7. It produces silk, but mainly for egg protection and making shelters, not for prey capture.
8. Respiration occurs through book lungs and/or tracheal system.
9. It exhibits sexual reproduction with internal fertilization.
10. Females lay eggs in silk egg sacs and show some degree of parental care.
11. It shows complex hunting behavior and strong vision-based predation, which is a key adaptive feature of salticid spiders.

2. Write a detailed note on life cycle and pathogenicity of *Ascaris lumbricoides*.

Life Cycle and Pathogenicity of *Ascaris lumbricoides*

Life Cycle

Ascaris lumbricoides is a human intestinal roundworm and follows a monoxenous life cycle (single host: humans).

1. ***Infective stage (Egg stage):*** The life cycle begins when a person ingests embryonated eggs through contaminated food, water, or hands.
2. ***Hatching in intestine:*** In the small intestine, the eggs hatch and release larvae.
3. ***Larval migration:*** The larvae penetrate the intestinal wall and enter the bloodstream. They travel through the liver → heart → lungs.
4. ***Pulmonary phase:*** In the lungs, larvae enter alveoli, grow, and then move up the respiratory tract to the throat.
5. ***Swallowing phase:*** The larvae are swallowed again and return to the small intestine.
6. ***Adult stage:*** In the small intestine, they mature into adult worms, mate, and produce eggs.
7. ***Egg release:*** Fertilized eggs are passed out with feces, contaminating soil and water. In the soil, they become infective after embryonation.

Pathogenicity (Disease Caused: Ascariasis)

1. ***Nutritional deficiency*:** Adult worms absorb nutrients, leading to malnutrition, weight loss, and stunted growth, especially in children.
2. ***Intestinal blockage:*** Heavy infections may cause intestinal obstruction, which can be fatal if untreated.
3. ***Tissue damage during migration:*** Larval movement through lungs can cause Löffler's syndrome, characterized by coughing, wheezing, and eosinophilia.
4. ***Abdominal pain and discomfort:*** Worms irritate the intestinal lining, causing cramps, nausea, and vomiting.
5. ***Biliary and pancreatic blockage:*** In severe cases, worms may enter bile ducts or pancreas, causing jaundice and inflammation.
6. ***Allergic reactions:*** Presence of worms may trigger immune responses and hypersensitivity reactions.
7. ***Poor growth and anemia:*** Long-term infection reduces nutrient absorption, leading to weakness and anemia in children.

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3. Write detailed characteristics of *Branchionus plicatilis*.

Characteristics of *Brachionus plicatilis*

1. *Brachionus plicatilis* is a microscopic rotifer belonging to class Monogononta and phylum Rotifera.
2. It is mainly a marine and brackish water organism, commonly found in estuaries, coastal lagoons, and salt ponds.
3. The body is bilaterally symmetrical, triploblastic, and pseudocoelomate, with organ system level of organization.
4. The body is divided into head, trunk, and foot regions.
5. The head bears a well-developed corona (ciliated wheel-like structure) used for locomotion and feeding by creating water currents.
6. It has a characteristic mastax with trophi, which helps in grinding microscopic food particles such as algae and bacteria.
7. The body is covered by a thin cuticle and may show a flexible, partially transparent appearance.
8. The foot region is short and ends in toes with adhesive glands used for temporary attachment to substrates.
9. Respiration occurs through the body surface by diffusion, as no specialized respiratory organs are present.
10. The excretory system is of protonephridial type**,** which also helps in osmoregulation in saline environments.
11. Reproduction is mainly parthenogenetic (asexual) under favorable conditions, but sexual reproduction occurs under stress.
12. Males are smaller and short-lived, appearing only during unfavorable conditions.
13. It plays an important ecological role as a primary consumer in aquatic food chains.
14. It is widely used in aquaculture as live feed for fish larvae due to its high nutritional value and rapid reproduction rate**.**

4. Write a detailed note on Dichotomous Key.

Dichotomous Key

A dichotomous key is a scientific tool used for the identification and classification of organisms based on a series of paired, contrasting statements. The word “dichotomous” means “divided into two parts,” which reflects the structure of the key.

It works by presenting the user with two choices at each step, where each choice leads either to the identification of the organism or to another pair of statements until the correct identification is reached.

A dichotomous key is mainly based on observable characteristics such as body structure, shape, size, presence or absence of specific features, habitat, or other morphological traits. There are two common types of dichotomous keys:

1. ***Indented key:*** In this type, each step is indented to show progression from general to specific features.
2. ***Bracketed key:*** In this type, paired statements are numbered and the correct choice directs the user to another number.

The construction of a dichotomous key involves selecting clear, contrasting, and non-overlapping characteristics to avoid confusion. Each step should be simple and based on easily observable traits.

Dichotomous keys are widely used in taxonomy, biology laboratories, ecological studies, and field identification of plants and animals. They help scientists and students quickly and accurately identify species without requiring advanced equipment.

They are especially important in studying biodiversity because they provide a systematic and logical method of classification**.**

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5. Write general characters of Nematodes.

General Characters of Nematodes

1. Nematodes are commonly known as roundworms and belong to phylum Nematoda.
2. They are triploblastic, bilaterally symmetrical, and unsegmented animals.
3. The body is long, cylindrical, and tapered at both ends, covered by a tough protective cuticle.
4. They possess a pseudocoelom (false body cavity) that acts as a hydrostatic skeleton.
5. The body shows organ-system level of organization, but true circulatory and respiratory systems are absent.
6. The digestive system is complete, with a mouth and anus.
7. They have a simple nervous system consisting of a nerve ring and longitudinal nerve cords.
8. Excretion is carried out by a specialized system of excretory canals or renette cells.
9. Respiration occurs through the body surface by diffusion.
10. Most nematodes are dioecious (sexes are separate), with males generally smaller than females.
11. Reproduction is mainly sexual, and fertilization is internal.
12. They may be free-living in soil and water or parasitic in plants and animals.
13. Some species are of great medical, agricultural, and veterinary importance due to their parasitic nature.

6. What do you know about economic importance of Earthworm?

Economic Importance of Earthworm

Earthworms are highly beneficial organisms that play an important role in soil ecology, agriculture, and environmental balance.

1. ***Soil fertility improvement:*** Earthworms break down organic matter and convert it into nutrient-rich humus, which increases soil fertility.
2. ***Soil aeration and mixing:*** By burrowing through the soil, they create channels that improve air circulation and water infiltration, enhancing root growth.
3. ***Vermicomposting:*** Earthworms are used in vermiculture to decompose organic waste and produce high-quality organic fertilizer called vermicompost.
4. ***Waste recycling:*** They help in the decomposition of agricultural waste, kitchen waste, and plant debris, reducing environmental pollution.
5. ***Improvement of soil structure:*** Their castings (excreta) improve soil texture, making it more loose, fertile, and better for cultivation.
6. ***Increase in crop yield:*** Soils rich in earthworms support better plant growth, resulting in higher agricultural productivity.
7. ***Indicator of soil health:*** Presence of earthworms indicates healthy and fertile soil conditions.
8. ***Use in fisheries and poultry feed:*** Earthworms are sometimes used as protein-rich feed for fish, birds, and other animals.
9. ***Medicinal and research use:*** Certain compounds derived from earthworms are studied for medical and biochemical research purposes.

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7. Write a detailed note on habitat of *Octopus vulgaris.*

Habitat of *Octopus vulgaris*

Octopus vulgaris (common octopus) is a marine cephalopod widely distributed in warm and temperate seas of the world. It is mainly found in coastal and shallow marine waters, although it may also occur in deeper regions depending on environmental conditions.

This species prefers habitats with rocky bottoms, coral reefs, and seagrass beds, where it can easily hide in crevices, holes, and caves. Such environments provide both protection from predators and suitable places for ambushing prey.

It is a benthic organism, meaning it lives on or near the sea floor. It is commonly found at depths ranging from the intertidal zone to about 200 meters, but most individuals inhabit relatively shallow waters. *Octopus vulgaris* is a solitary and highly territorial animal, often occupying a fixed den that it modifies using shells, rocks, and other materials. It shows strong site fidelity and returns to its shelter after hunting.

It prefers areas with clear water and abundant food availability, such as crustaceans, molluscs, and small fishes. It is highly adaptable and can survive in a wide range of marine environmental conditions, provided there is sufficient shelter and food supply.

8. Write a detailed note on diversity and habitat of Echinoderms.

Diversity and Habitat of Echinoderms

Diversity of Echinoderms

Phylum Echinodermata includes exclusively marine invertebrates with a high level of diversity in body form and function. The phylum is divided into five major classes:

1. ***Asteroidea (sea stars)*** – star-shaped body with arms; mainly predators feeding on molluscs. ***Example:*** Asterias
2. ***Ophiuroidea (brittle stars)*** – have slender, flexible arms distinct from central disc; move quickly using arms. ***Example:*** Ophiothrix
3. ***Echinoidea (sea urchins and sand dollars)*** – globular or flattened body without arms; covered with spines. ***Example:*** Echinus
4. ***Holothuroidea (sea cucumbers)*** – elongated, soft-bodied echinoderms with reduced skeleton. *Example:* Holothuria
5. ***Crinoidea (sea lilies and feather stars)*** – sessile or free-moving forms with feathery arms used for filter feeding. ***Example:*** Antedon

Habitat of Echinoderms

Echinoderms are exclusively marine organisms and are found in all oceans of the world, from shallow coastal waters to deep sea regions.

- Most species live on the sea floor (benthic zone).
- They are commonly found in habitats such as coral reefs, rocky shores, sandy beaches, and ocean depths.
- Some species like sea cucumbers inhabit muddy or sandy sediments, where they play an important role in nutrient recycling.
- Crinoids may attach to rocks or corals in deep marine environments, while brittle stars often hide under stones or within crevices.

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9. What are High Weight Characters?

High Weight Characters

High weight characters are those taxonomic or morphological characters that carry greater importance or significance in classification and identification of organisms compared to other characters.

These characters are considered more reliable because they are stable, less influenced by environmental changes, and genetically controlled**.**

High weight characters are mainly used in systematics and taxonomy to distinguish major groups such as phyla, classes, and orders. Examples of high weight characters include:

- Body symmetry (radial or bilateral)
- Presence or absence of coelom
- Type of body cavity (true coelom, pseudocoelom, or none)
- Segmentation of body
- Level of organization (cellular, tissue, organ-system)
- Type of skeleton (endoskeleton or exoskeleton)

Such characters are more important than low weight characters, which include features like color, size, and habitat, as these may vary within species or due to environmental conditions.

10. Write significance of Dichotomous key.

Significance of Dichotomous Key

1. A dichotomous key helps in the easy and systematic identification of organisms based on their observable characteristics.
2. It provides a logical and step-by-step method, making identification simple even for beginners.
3. It is widely used in taxonomy and systematics for classifying plants, animals, and microorganisms.
4. It helps in distinguishing closely related species by comparing contrasting characteristics.
5. It reduces confusion by using clear paired choices (either/or statements) at each step.
6. It is useful in field studies and laboratory work where quick identification is required.
7. It promotes accurate scientific classification and reduces errors in identifying organisms.
8. It is an important tool in biodiversity studies and ecological research.
9. It helps students develop observational and analytical skills in biology.

11. Write significance of F-test in ANOVA.

Significance of F-test in ANOVA

1. The F-test is the core statistical test used in ANOVA to compare variances between groups.
2. It helps determine whether there is a significant difference among the means of three or more groups.
3. The F-value is calculated as the ratio of between-group variance to within-group variance.
4. A high F-value indicates that the variation between group means is greater than the variation within groups, suggesting a real difference among treatments.
5. It allows researchers to test the null hypothesis, which states that all group means are equal.
6. If the calculated F-value is greater than the critical value, the null hypothesis is rejected, showing statistically significant differences.
7. The F-test is essential in experimental research, biology, agriculture, psychology, and other sciences for comparing multiple groups simultaneously.
8. It provides a more efficient method than multiple t-tests, reducing the risk of Type I error (false positives)**.**

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12. Define the structure and body size of annelids.

Structure and Body Size of Annelids

Structure of Annelids:

Annelids are triploblastic, bilaterally symmetrical, and coelomate animals. Their body shows true segmentation (metamerism), where the body is divided into a series of repeated segments called metameres. Each segment is separated by internal septa and may contain repeated organs.

The body is covered by a thin, flexible cuticle, and the body wall contains circular and longitudinal muscles that help in movement. Most annelids possess setae (bristles) for locomotion, while parapodia are present in some groups like polychaetes. The body is organized into regions such as head, trunk, and sometimes a terminal pygidium.

They have a well-developed coelom filled with coelomic fluid, which acts as a hydrostatic skeleton and assists in movement and transport of substances.

Body Size of Annelids:

Annelids show a wide range in body size. They may be very small, microscopic forms, or they may grow into large-sized organisms.

- Some small annelids measure only a few millimeters in length.
- Earthworms (e.g., Pheretima) are typically several centimeters long.
- Some marine polychaetes (e.g., Nereis) may also reach moderate sizes.
- Large earthworms in tropical regions can grow up to 1 meter or more in length.

13. Write a detailed note in T-test.

T-test

The t-test is a statistical test used to determine whether there is a significant difference between the means of two groups. It is commonly used in biological, medical, and experimental studies where sample sizes are small and population variance is unknown. The t-test is based on the t-distribution, which is similar to the normal distribution but has heavier tails, making it suitable for small samples.

Types of t-test

1. ***Independent (Unpaired) t-test:*** This is used when two different groups are compared. ***Example***: Comparing plant growth under two different fertilizers applied to separate groups.
2. ***Paired t-test:*** This is used when the same group is measured twice under different conditions. ***Example***: Measuring blood pressure of patients before and after treatment.
3. ***One-sample t-test:*** This is used to compare the mean of a single sample against a known population mean.

Formula (basic idea)

The t-value is calculated as: t = (Difference between sample means) / (Standard error of difference)
A higher t-value indicates a greater difference between group means.

Assumptions of t-test

- Data should be normally distributed
- Samples should be independent (except paired t-test)
- Data should be measured on interval or ratio scale
- Variances should be approximately equal (for independent t-test)

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Applications

- Biology experiments (growth studies, enzyme activity)
- Medical research (treatment effectiveness)
- Agriculture (fertilizer comparisons)
- Psychology and social sciences (behavioral studies)

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